

Aniket Agarwal

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Experience

Gatik

Machine Learning Engineer | Autonomy & Robotics

Aug 2025 – Present
Mountain View, California

- Spearheaded end-to-end model development for the production obstacle detection pipeline, driving a **+20 mAP improvement on underrepresented object classes** (buses, cyclists) and **+12 mAP overall** through a hybrid sparse-convolutional (spconv) and transformer-based 3D detection architecture deployed in active fleet testing.
- Developing a diffusion-based imitation learning pipeline with post-trained RL fine-tuning for a learned motion planner, targeting improved generalization and robustness in long-tail and safety-critical scenarios for closed-loop fleet deployment.
- Led the development of an implicit query-based 3D occupancy prediction network for scalable scene representation, providing robust obstacle coverage for rare and out-of-distribution objects that evade explicit detection-based classification.

Carnegie Mellon University

Research Assistant | Advisor: Prof Laszlo Jeni

Nov 2023 – July 2025
Pittsburgh, Pennsylvania

- **MSR Thesis:** Developed a multimodal HMR framework fusing RGB, LiDAR, and IR data via a transformer-based Modality Unifier for robust 3D pose estimation under degraded sensing, achieving *14% lower MPJPE* than single-modality baselines; deployed on Jetson Orin for real-world casualty assessment. Report
- Built multimodal perception system for **DARPA Triage Challenge** integrating RGB, thermal, and audio streams; fine-tuned and quantized LLM/TTS models for edge deployment on NVIDIA Jetson with real-time ROS inference pipelines.

Microsoft

Data & Applied Scientist | MSAI Team

July 2022 – Aug 2023
Hyderabad, India

- Built and deployed Learning-to-Rank models serving **100M+ daily queries** for Outlook ranking, improving NDCG@10 by 10% and user engagement by 5%; reduced model latency by 20% via feature importance-based dimensionality reduction.

Princeton University

Research Intern | Advisor: Prof Karthik Narasimhan

Oct. 2021 – Jun 2022
New Jersey, USA

- Created and released **LVU-Sports benchmark** for long-form video understanding, processing 10K+ hours of footage with temporal action detection; human evaluation revealed 70% performance gaps in existing models, enabling new research directions. DMLR@ICLR 2024

Publications

1. **Aniket Agarwal**, Laszlo A Jeni. “Multimodal Human Mesh Recovery for Stand-off Triage in Mass Casualty Scenarios” *CMU MSR Thesis — In Submission at NeurIPS 2026* [Project Page]
2. **Aniket Agarwal***, Alex Zhang*, Karthik Narasimhan, Igor Gilitschenski, Vishvak Murahari, Yash Kant. “Building Scalable Video Understanding Benchmarks through Sports.” *Data-Centric ML Research Workshop at International Conference on Learning Representations (DMLR@ICLR), 2024.* [Project Page]
3. Jun Chen, **Aniket Agarwal**, Sherif Abdelkarim, Deyao Zhu, Mohamed Elhoseiny. “RelTransformer: A Transformer-Based Long-Tail Visual Relationship Recognition.” *Computer Vision and Pattern Recognition Conference (CVPR), 2022.* [Paper Link]
4. **Aniket Agarwal***, Sherif Abdelkarim*, Panos Achlioptas, Jun Chen, Jiaji Huang, Boyang Li, Kenneth Church, Mohamed Elhoseiny. “Exploring Long Tail Visual Relationship Recognition with Large Vocabulary.” *International Conference on Computer Vision (ICCV), 2021.* [Project Page]

Education

Carnegie Mellon University - School of Computer Science

Master of Science in Robotics | **CGPA: 3.90/4.00**

Pittsburgh, PA
Aug. 2023 - July 2025

Indian Institute of Technology Roorkee

Integrated Master of Science in Applied Mathematics | **CGPA: 7.82/10**

Roorkee, India
July 2017 – May 2022

Technical Skills

Research Interests: Multi-modal perception, 3D Computer Vision, efficient neural representations, large-scale dataset curation, Vision-Language Models, Autonomous Systems

Frameworks/Libraries: PyTorch, PyTorch Lightning, PyTorch3D, Huggingface, PCL, TensorRT, ONNX, ROS2, OpenCV

Languages: Python, C++, CUDA, Bash